

# Companion XXXII: Halo Concentration, Density Profiles, and Substructure in the Relaxation-Driven Cyclic Cosmology

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## Abstract

This Companion investigates how the Relaxation-Driven Cyclic Cosmology (RDCC) modifies the internal structure of dark matter halos. While Companion XXXI established the non-linear halo model and its global implications, the present work focuses on the microstructure of halos: their density profiles, concentration–mass relation, subhalo abundance, and the emergence of core-like features. The relaxation-induced suppression of small-scale potential evolution alters the collapse history of halos and reshapes their internal architecture. We derive analytical estimates, discuss scaling relations, and outline observational signatures relevant for weak lensing, galaxy–galaxy lensing, and cluster astrophysics.

## 1 Introduction

Dark matter halos are not only the building blocks of cosmic structure but also repositories of dynamical information. Their internal density profiles, concentration parameters, and substructure content encode the history of gravitational collapse. In the standard cosmological model, halo structure is shaped by hierarchical growth and the scale-free nature of cold dark matter.

In the Relaxation-Driven Cyclic Cosmology (RDCC), the situation changes fundamentally. The relaxation-modified potential evolution introduces a scale-dependent suppression of small-scale growth, which alters the timing and efficiency of halo collapse. As a result, halo profiles, concentrations, and subhalo populations acquire characteristic RDCC signatures.

This Companion develops the theoretical framework for halo microstructure in RDCC.

## 2 Density Profiles in RDCC

The standard Navarro–Frenk–White (NFW) profile is

$$\rho_{\text{NFW}}(r) = \frac{\rho_s}{(r/r_s)(1 + r/r_s)^2}. \quad (1)$$

In RDCC, the suppression of small-scale potential evolution modifies the inner collapse dynamics. We model the RDCC profile as

$$\rho_{\text{RDCC}}(r) = \rho_{\text{NFW}}(r) \left[ 1 - \chi_\rho \left( \frac{r}{r_{\text{rel}}} \right)^\beta \right], \quad (2)$$

where  $r_{\text{rel}}$  is the scale corresponding to  $k_{\text{rel}}$ .

RDCC predicts:

- slightly shallower inner slopes,
- smoother transitions between inner and outer regions,
- reduced central densities for low-mass halos,
- preserved outer profiles for massive halos.

### 3 Concentration–Mass Relation

The concentration parameter is defined as

$$c(M) = \frac{r_{\text{vir}}}{r_s}. \quad (3)$$

In RDCC, the delayed collapse of small halos and the earlier collapse of massive halos modify the concentration–mass relation:

$$c_{\text{RDCC}}(M) = c(M) \left[ 1 - \chi_c \left( \frac{k(M)}{k_{\text{rel}}} \right)^\alpha \right]. \quad (4)$$

Consequences:

- reduced concentrations for dwarf-scale halos,
- nearly unchanged concentrations for cluster-scale halos,
- a characteristic turnover mass  $M_{\text{rel}}$ ,
- smoother redshift evolution of  $c(M)$ .

### 4 Subhalo Abundance and Survival

Subhalos trace the hierarchical assembly of halos. Their abundance is commonly described by the subhalo mass function:

$$\frac{dN}{d \ln m} \propto m^{-\gamma}. \quad (5)$$

In RDCC, the suppression of small-scale growth reduces the formation of low-mass subhalos:

$$\left( \frac{dN}{d \ln m} \right)_{\text{RDCC}} = \left( \frac{dN}{d \ln m} \right) \left[ 1 - \chi_{\text{sub}} \left( \frac{k(m)}{k_{\text{rel}}} \right)^\alpha \right]. \quad (6)$$

RDCC predicts:

- fewer low-mass subhalos,
- enhanced survival of intermediate-mass subhalos,
- reduced tidal stripping due to smoother potentials,
- a natural explanation for missing-satellite-like features.

## 5 Core Formation and Inner Structure

The relaxation-modified potential evolution leads to a reduced efficiency of small-scale collapse. This can produce core-like features in low-mass halos.

We model the RDCC core radius as

$$r_{\text{core}} \sim r_s \left( \frac{k_{\text{rel}}}{k(M)} \right)^\alpha. \quad (7)$$

Implications:

- dwarf-scale halos develop shallow cores,
- Milky-Way-scale halos remain cusp-like,
- cluster-scale halos are unaffected,
- core sizes correlate with  $k_{\text{rel}}$ .

## 6 Observational Signatures

The RDCC modifications to halo structure affect several observables:

- **Galaxy–galaxy lensing:** reduced small-scale shear signals.
- **Weak lensing peaks:** fewer high-concentration peaks.
- **Rotation curves:** natural emergence of shallow cores.
- **Cluster lensing:** preserved NFW-like outer profiles.
- **Substructure lensing:** reduced subhalo-induced perturbations.

These signatures provide a direct test of RDCC on non-linear scales.

## 7 Conclusion

This Companion develops the internal structure of dark matter halos in the Relaxation-Driven Cyclic Cosmology. The relaxation-modified potential evolution reshapes halo profiles, concentrations, and subhalo populations. These effects propagate into lensing, galaxy dynamics, and the morphology of the cosmic web. Together with Companion XXXI, the present work establishes the foundation for RDCC non-linear cosmology and prepares the ground for future studies of galaxy formation, cluster astrophysics, and cosmic web topology.

## References

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